21 August 2024

Lecture 2

Introduction to thermodynamics

Energy transfer heat, work and mass

Thermodynamics deals with the transfer of energy from one form to another or from one place to another

Energy A quantitative property that must be transferred from a body or physical system to perform work or transfer heat

Heat Form of energy that is transfered between two bodies due to difference in the temperature.

Forms of Energy

1. Kinetic energy
2. Potential energy
3. Thermal energy
4. Chemical energy
5. Nuclear energy
6. Electrical energy
7. Electromagnetism

energy is always conserv.

Work the amount of force required to move a body with constant acceleration

Form of energy

- Thermal energy
- Kinetic energy
- potential energy
- chemical energy

Temperature
Heat] difference

Further Classification of Energy:

Energy

Macroscopic energy
↳ Kinetic Energy
↳ Potential energy

Microscopic energy
↳ Electrical energy
↓

Internal Energy (U)
- Translational energy
- Vibrational energy
- Rotational energy
- Binding forces

Total energy = Macroscopic + Microscopic

Total Energy = PE + KE + U

$$\text{Total Energy} = mgh + \frac{1}{2}mv^2 + U$$

Mechanism of Energy Transfer

1. Work done (W) ⟶ Joule → Mechanical Energy
2. Heat (q) ⟶ Joule → Thermal energy
3. Mass flow

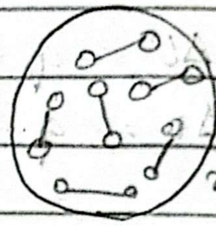
Surroundings

⟶ −300J

Heat ⟶ | System |

$\Delta U = +300J$ ↑ work +300J

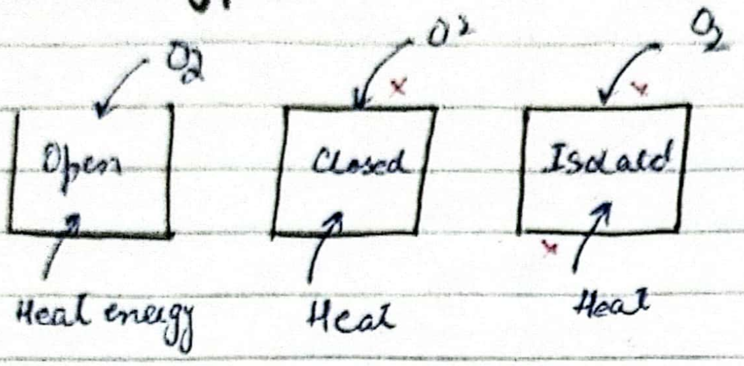
* Macroscopic large am of energy required
* Electrical energy is microscopic because depend on amount of current
* Most commonly KE and PE is macroscopic. Molecule have charge which is continuously moving and they have non linear motion moves rand



m a

* As it is difficult locate microscopic energy individually is difficult ~~So we name all~~ internal energy
* The sum of micr and macroscopic is total energy.
* heat move from high ~~temperature to~~ l temperature
* Internal Energy, work must have signs
*

Systems types:



$$\Delta U = q + w \quad \text{(Chemistry)}$$

$$\boxed{\Delta U = q - w \quad \text{(Physics)}}$$

~~W = −ve → work done By the system~~
~~W = + ve → work done ON the system~~] Chemistry

W = − ve → work done ON the system
W = + Ve → work done By the system] Physics

open system (water [illegible] surrounding se some sort of energy heat and [illegible] [illegible] [illegible] [illegible])
Close System (water [illegible] mass flow nahi hai [illegible] heat transfer ho raha surroundings is effecting)
Isolated System
No heat transfer no mass flow

* Surrounding se energy ly rahay hai to [illegible] usko negative bolege hamari positive hai
* we need to follow physics expressions

Physics ma aur [illegible] ka prespective se hota hai Chemistry ma Surroundings consider nahi hota

Heat

$q \longrightarrow$ +ve (absorb) $\longrightarrow$ endothermic
$q \longrightarrow$ −ve (release) $\longrightarrow$ exothermic

where heat is release exothermic
exo → outside
where heat is abs endothermic
endo → inside

Example 100 J of work is done on the system U increases by 74 J How much energy is transferred as heat?

$w = -100$ (as on the system) $U = 74\,J$

$$\Delta U = q - w$$
$$74 = q - (-100)$$
$$74 = q + 100$$
$$\boxed{q = -26\,J}$$

q represent heat
−26 heat is decrease from the system
heat is lost

Heat is lost and it is exothermic

Example A sample of gas does 150 J of work against its surroundings and loses 90 J of internal energy in the process? Does the gas lose or gain the heat and how much?

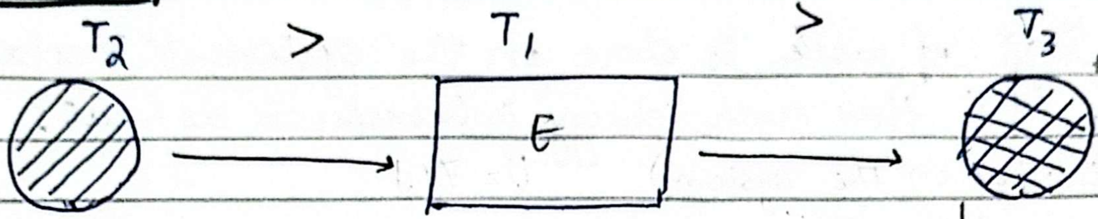
work done by the system

$w = 150$ $U = -90$ (as it is lost)

$$\Delta U = q - w$$
$$-90 = q - 150$$
$$-90 + 150 = q$$
$$\boxed{q = +60\,J}$$

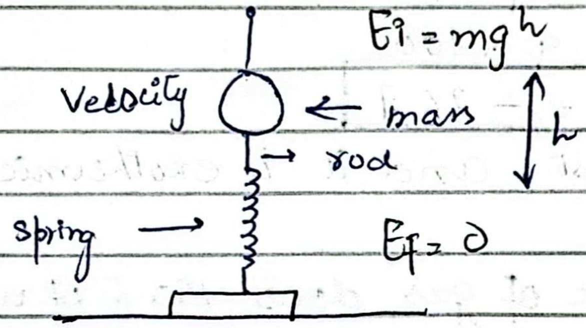
Heat is gain and it is endothermic

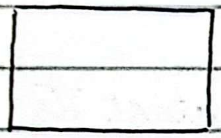
Heat transfer



* Heat move from high temperature to low
$T_2 > T_1$
so move from T_2 to T_1
and
$T_1 > T_3$
so move from T_1 to T_3
Example
[Sunlight
Fire

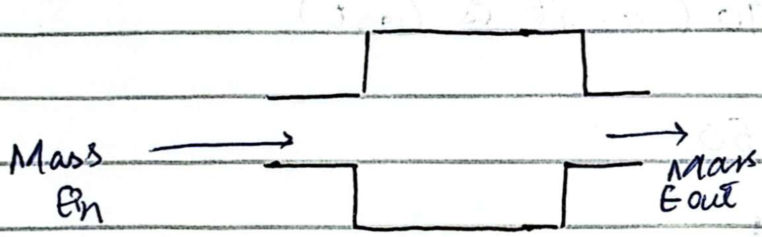
Work





* As we move the mass to height so it is potential energy which is mgh when we leave back it become zero
As it move downward there is frictional energy, sound energy, heat energy as it oscilates so energy become zero.

Mass flow



* Mass have some sort of energy anything created have some sort of energy as energy can neither be created nor be destroyed

28 August 2024

Lecture 3

Properties of Pure Substance

Pure substance A pure substance is define as a material that has a homogenous and invariable chemical composition.

Example H_2O, CO_2

Homogenity The substance is the same throughout with no regions having a different composition

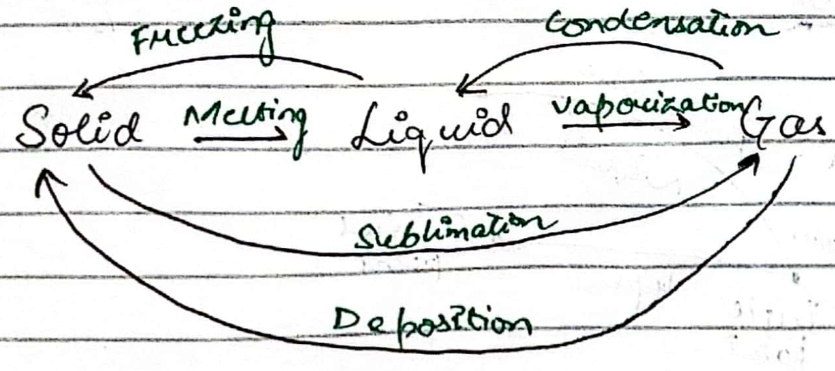
Invariable Composition The chemical composition does not change even when the substance, transition between different phases

Not pure substance
↳ Rubber
↳ Methane
↳ Salt

Types of phases
Solid Tightly packed molecules
Liquid Tightly packed but movable
Gas Freely movable

Phase Changes and Diagrams

Phase changes of a material occurs when it transition from one phase/state to another

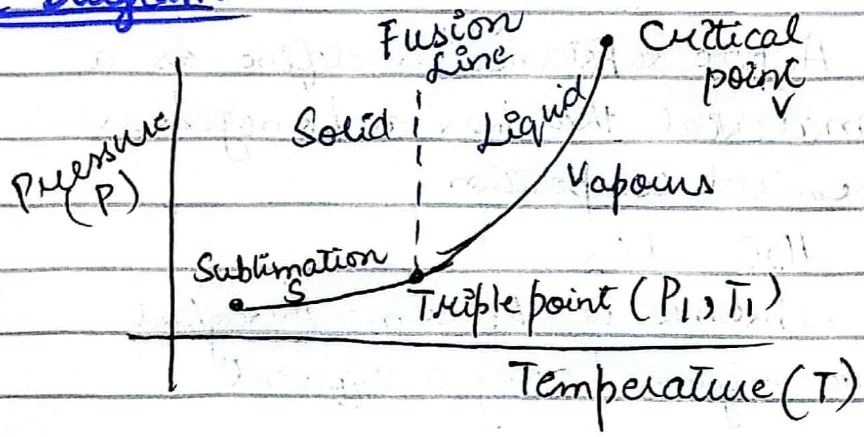


water phase diagram

pressure ↓ temperature ↑

Saturation Temperature The specific temperature at which a substance undergoes a phase change under a given pressure (vaporization, condensation)

Phase Diagram



* Fusion Line (separate solid liquid state)
* Liquid sa gas vaporization
* Sublimation

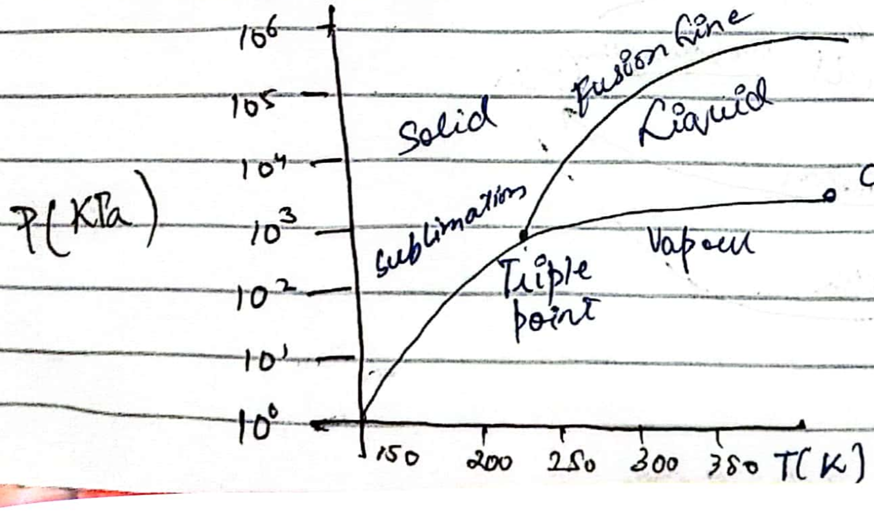
* Triple point (all three states of material exist)
* Critical point gaseous & liquid state no difference

Triple point of water occurs at 0.01°C, 0.6117KPa

Critical point marks at the end of vaporization curve on a phase diagram Beyond this point Liquid and gaseous phase becomes indistinguishable.

Critical point of water : 374.14°C and 22.09 MPa

Phase diagram of CO_2



* Fusion line b/w solid & liquid hoti
* Critical point Liquid & vapour ka b/w hoti
* Sublimation solid sa liquid

Q What is the lowest temperature at which water can remain in the Liquid form?

0.01

at 0.01 °C
it is
triple point
water is in
both solid &
Liquid state

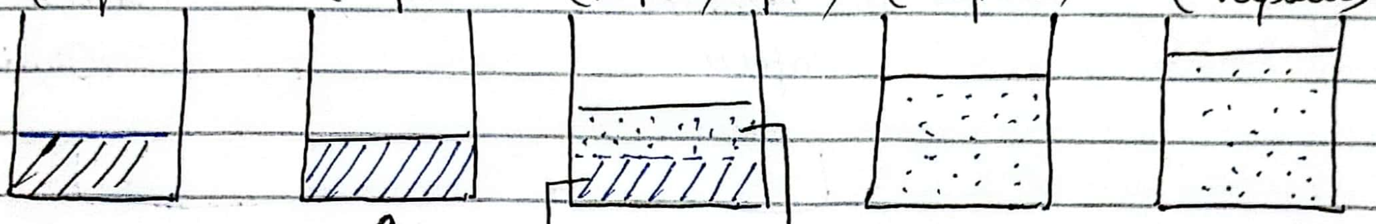
Q If the *(smaller)* pressure is smaller than the smallest pressure P_{sat} at a given temperature T. What is the phase?

4 Sep 2024

Lecture 5

Tables of thermodynamics Properties

State 1 (Liquid) | State 2 (Liquid) | State 3 (Liquid/vapour) | State 4 (Vapour) | State 5 (Vapour)



Heat

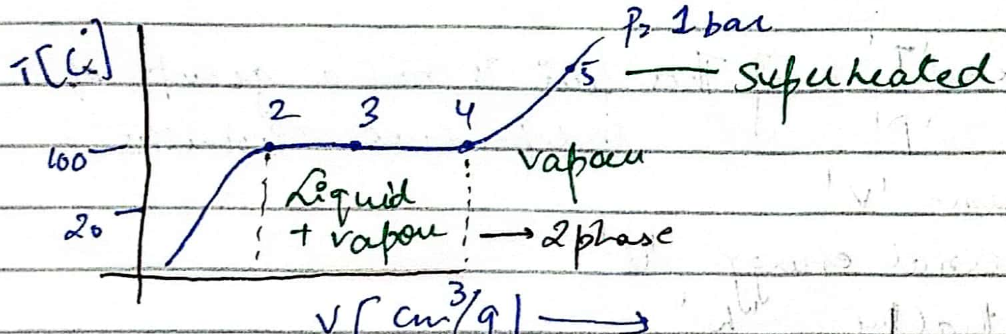
Vapour
Liquid

State 1: $P = 1$ bar, $T = 20°C$ → Compressed Liquid

State 2: $P = 1$ bar, $T = 100°$

State 3: $P = 1$ bar, $T = 100°C$

State 5: $P = 1$ bar, $T = 100°C$ → Superheated Vapour



Purpose

Thermodynamics tables provide essential data for various substances that describe three thermodynamic property (like temperature, pressure, specific volume, internal energy, enthalpy, entropy) mixture

Compress Liquid	Saturated Liquid	Saturated Liquid Vapour mixture	Saturated Vapour
Table 1	Liquid about to become vapour	Temp & pressure actually dependent on each other	Vapour about to become Liquid

1. Pressure (Table No-2)
2. Temperature (Table No.3)

With two independent properties, we can find the rest

- Temperature 'T'
- Pressure 'P'

This pair is not a group of independent properties

- Specific volume 'v'
- Specific Internal energy 'U'
- Specific Enthalpy 'h'
- Specific energy 'S'

Satura
Supperheated Vapour
Table 4

Saturation Temperature (Tsat) :

Temperature at which phase change occur at constant Pressure

Saturation Pressure (Psat)

Pressure at which phase change occur at constant temperature

T	P	V_f	V_{fg}	V_g
0	P_1	V_1	V_{11}	V_{g1}
10	P_2	V_2	V_{12}	V_{g2}
20	P_3	V_3	V_{13}	V_{g3}
30	P_4	V_4	V_{14}	V_{g4}
40	P_5	V_5	V_{15}	V_{g4}

- Linear Interpolation
- Polynomial regression
- Moving average
- Language Interpolation

Saturate Liquid and Vapour Tables

Table B 1.2

V_f: Specific volume of saturated Liquid

V_g: Specific Volume of the saturated vapour

V_{fg}: Difference b/w V_g and V_f

$$V_{fg} = V_g - V_f$$

First column : Temperature

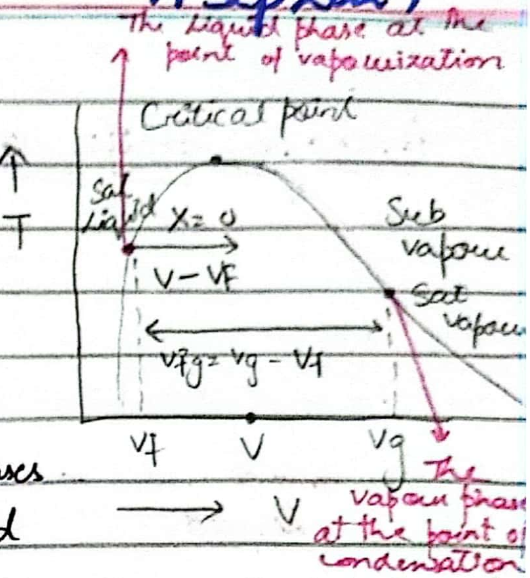
Second column: Corresponding Saturation Pressure (Ps

Table B1.2

- Contains the same information as Table B1.1 but organized pressure
- Useful when pressure is known and Temperature needs to be found.

11 Sep 2024

Lecture 6



$v_f \rightarrow$ saturated Liquid

The two phase States

Two Phase State:

A mixture of two phases of the same substance, such as liquid and vapour in equilibrium at the same temperature (T) and pressure (P)

Specific Volume:

Volume occupied by a unit mass of a substance

Measure of how much space one kg of substance occupies

If total volume is given than divide by its mass for specific volume

$$v = \frac{V}{m} \Rightarrow \boxed{V = mv}$$

Specific Volume in Two phase Mixture:

Total Volume of two phase mixture is the sum of the volume of the Liquid (V_{Liq}) and Vapour (V_{vap})

$$V_{Total} = V_{Liq} + V_{vap}$$
$$= m_{liq} v_f + m_{vap} v_g$$

Average Specific Volume:

$$v = \frac{V}{m}$$

$$v = \frac{m_{Liq} v_f + m_{vap} v_g}{m}$$

mass of Liquid

$$v = \frac{m_{Liq} v_f}{m} + \frac{m_{vap} v_g}{m}$$

mass of vapour

$m \rightarrow$ whole mass of substance

$\because \frac{m_{Liq}}{m} = 1-x \qquad \frac{m_{vap}}{m} = x$ — Quality

$$v = \frac{m_{Liq} v_f}{m} + \frac{m_{vap} v_g}{m}$$

$$\boxed{v = (1-x) v_f + x v_g}$$

$$v = (1-x) v_f + x v_g$$

$$= v_f - x v_f + x v_g$$

$$= v_f + x (v_g - v_f) \qquad \because v_{fg} = v_g - v_f$$

$$\boxed{v = v_f + x v_{fg}}$$

Quality:

It is the ratio of mass of vapour to the total mass in a two phase mixture.

- Range $0 \leq x \leq 1$

If $x=0$ (saturated Liquid)
$x=1$ (saturated vapour)

Problem:

Calculate the specific volume of a saturated mixture of water at 200°C with a quality of 70%.
Refer Table B1.1

$$v = v_f + x\, v_{fg}$$

$v_f = 0.001156$ $v_{fg} = 0.12620$ $v_g = 0.12736$

$$v = v_f + x\,(v_g - v_f)$$
$$= 0.001156 + 0.7\,(0.12736 - 0.001156)$$
$$v = \underline{0.089498}8 \; m^3/Kg$$

4 decimals

Compressed Liquid:

A Liquid state at a pressure higher than the saturation pressure for a given temperature OR at a temperature lower than the saturation temp for a given pressure

Table B.1.4

$P > P_{sat}$ at a given T
$T < T_{sat}$ at a given P
$v < v_f$ at a given T or P

$u < u_f \rightarrow$ Internal energy

$h < h_f \rightarrow$ Enthalpy

Superheated Vapour:

A Liquid state at a pressure lower than the saturation pressure for a given temperature OR at a temperature higher than the saturation Temperature for a given pressure

$P < P_{sat}$ at a given T

$T > T_{sat}$ at a given P

$V > V_g$ at a given T or P

Problem 2.25

Determine the phase of water at

a) $T = 260°C$, $P = 5MPa$

b) $T = -2°C$ $P = 100 kPa$

a) we see temperature entry of water B 1-1

$T = 260°$ $P_{sat} = 4688.6 kPa$

$= 4.688 MPa$

$P > P_{sat}$

$5MPa > 4.688 MPa$

it is a compressed Liquid.

at $P = 5MPa = 5000 kPa$ Table B 1.2

$T_{sat} = 263.99 °C$

$T < T_{sat}$

$260 < 263.99 °C$

b) $P = 100 kPa$ $T = -2°C$

At $P = 100 kPa$ $T_{sat} = 99.62$

$T < T_{sat}$

$-2 < 99.62$

Table 1.5

At $T = -2°C$ $P_{sat} = 0.5177 kPa$

$P > P_{sat}$

$100 kPa > 0.5177$ compressed Solid

Example 2.1 water state

a) 120°C 500kPa

at 120°C

$P_{sat} = 198.5\ kPa$

$P > P_{sat}$

$500\ kPa > 198.5\ kPa$

500kPa

$T_{sat} = 151.86\ °C$

$T < T_{sat}$

$120° < 151.86$

compressed Liquid

b) 120°C 0.5m

16 Sep 2024

Lecture 7

Ideal Gas Law

Describes the behaviour of gases under certain conditions It provides simple correlation among pressure temperature (T) and specific volume (V) It is an approximation and works best under high temperature, low density and low pressure

Ideal Gas Equation of State

$$PV = RT$$

For calculations involving mass / moles

$$PV = mRT = nRT$$

$$R = \frac{\bar{R}}{M}$$ $\bar{R} \rightarrow$ constant, $M \rightarrow$ Moles

$$\bar{R} = 8.3145 \frac{KJ}{Kmol\,K}$$

$\therefore n = \frac{m}{M}$

$$PV = mRT = n\bar{R}T$$

n is the number of moles

Ideal Gas
- Pressure ↓
- Temperature ↑

Ideal Gas Law hold when
- Low pressure
- High temperature
- Low density

Mass flow rate $\rightarrow$ ideal gas Law

$P\dot{V} = \dot{m}RT = \dot{n}\bar{R}T$

$\cdot \longrightarrow$ derivative

flow $\rightarrow$ Kg/s²

Table A-5

Q/ Mass of air?

Room dimensions: 6m x 10m x 4m

Temp: 25°C , P = 100 KPa , Molecular Mass = 28.97 Kg/Kmol

$R = 0.287 \, KJ/Kg\,K$

$PV = mRT$

$(100\,kPa)(6 \times 10 \times 4) = m(\quad)(25)$

$$\frac{(100\,KPa)(6 \times 10 \times 4)}{(25+273)(0.287)} = m$$

$$\boxed{m = 280.6\,Kg}$$

$$R = \frac{\bar{R}}{M}$$

$$R = \frac{8.3145}{28.97}$$

$$\boxed{R = 0.287}$$

Q/ Tank has a volume: 0.5m³

m = 10 kg of ideal gas

molecular mass: M = 24 Kg/KmolK

T = 25°C

P = ?

$$R = \frac{\bar{R}}{M} = \frac{8.3145}{24}$$

$R = 0.3464$

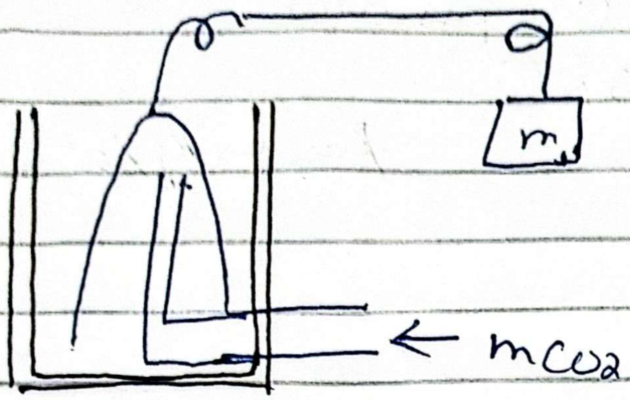
$$P = \frac{mRT}{V}$$

$$= \frac{10(0.3464)(25+273)}{0.5}$$

$$\boxed{P = 2064.5\,KPa}$$

CamScanner

Example 2.8 Pag



A gas bell is submerged within mass
The pressure is 105 KPa and temperature is 21°C.
a volume increase is measure to be $0.75 m^3$ over a period
185 sec What is the volume flow rate and mass
flow rate assuming it is carbon dioxide

$R = 0.1889\ KJ/Kg \cdot K$

$T = 21°C = 273 + 21 = 294 K$

$V = 0.75 m^3$

$P = 105 KPa$

$\dot{V} = \frac{\Delta V}{\Delta t} = \frac{0.75 m^3}{185 sec} = 0.00405\ m^3/s$

$\longrightarrow$ mass flow rate.

$P\dot{V} = \dot{m} RT$

↳ volume flow rate

$\dot{m} = \frac{P\dot{V}}{RT}$

$= \frac{105 \times 0.00405}{(0.1889)(294)} = 0.00765\ Kg/s$